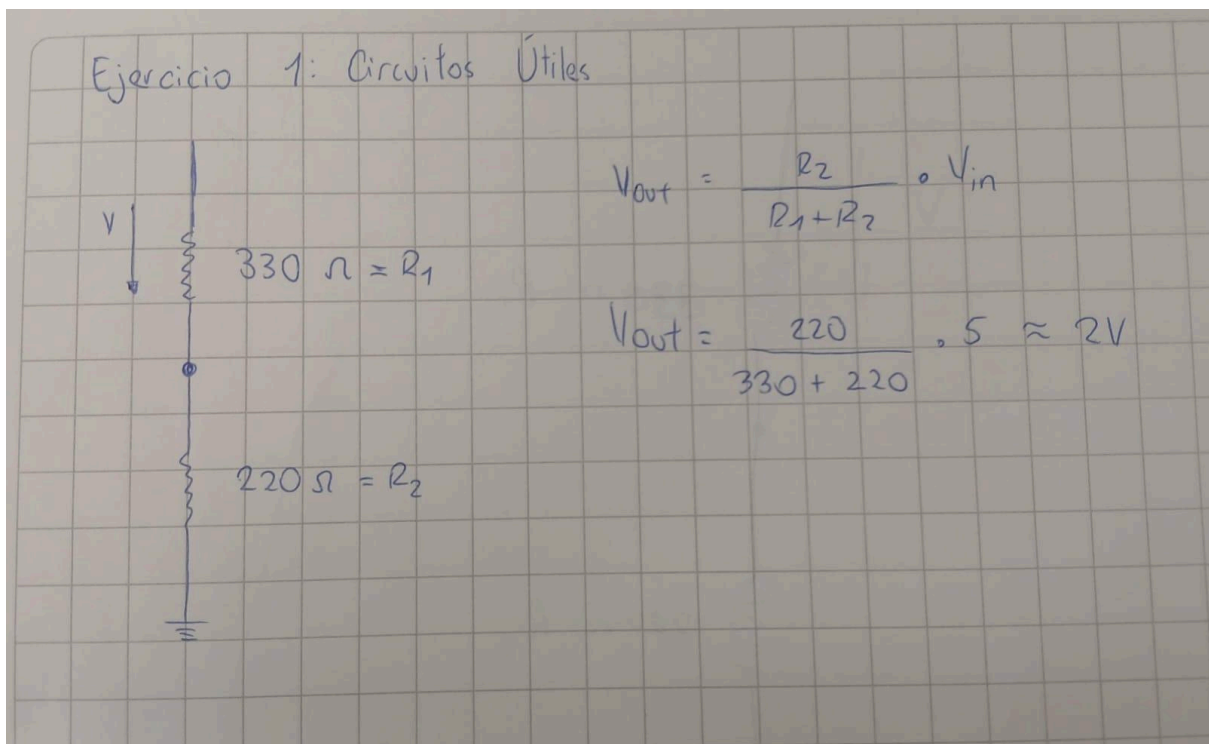
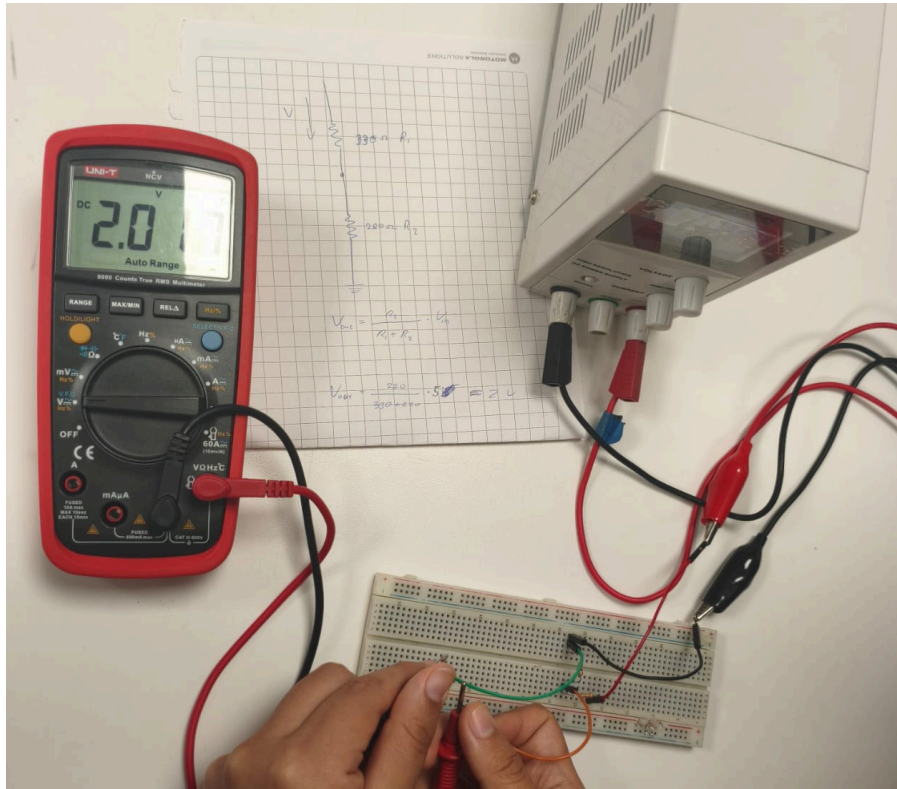


### Ejercicio 1:

Se observa la medición de los voltios, el sistema ya ensamblado con la fuente de poder a 5V y 1,77A



## Ejercicio 2:

Cálculo de Rangos de Resistencia para cada led

1) LED ROJO ( $V_{\text{lumbral}} = 1,8V$ )

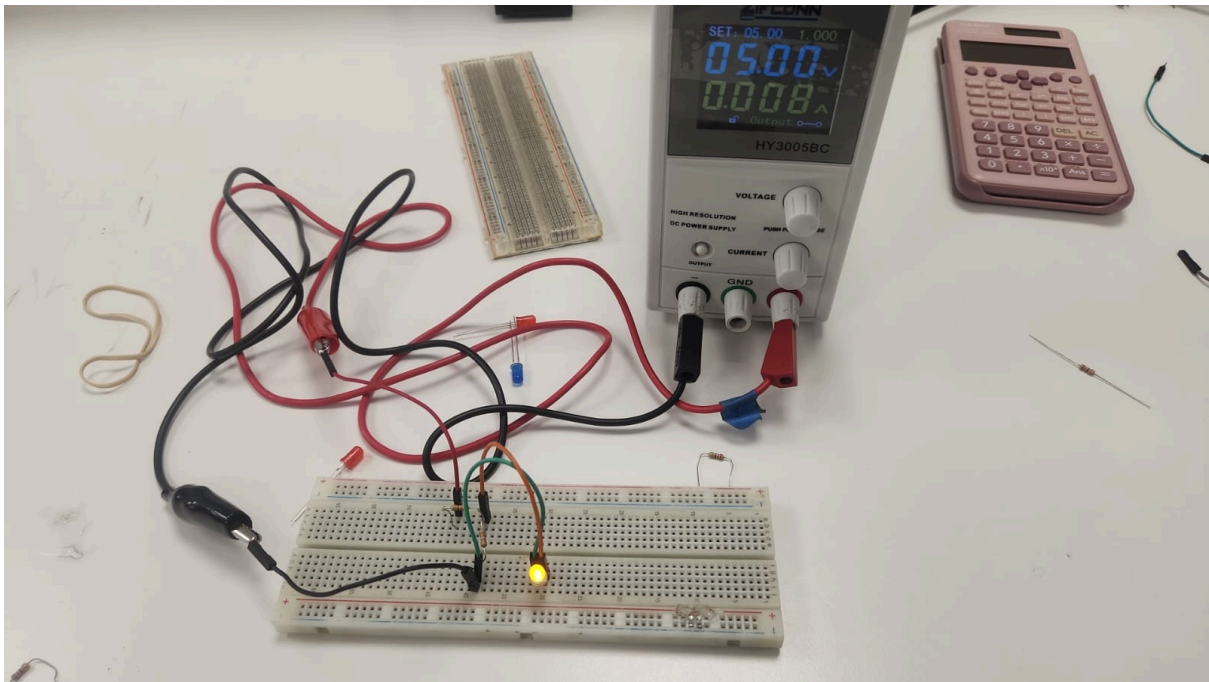
- $R_{\text{minima}} = \frac{5V - 1,8V}{0,01A} = 320\Omega$
- $R_{\text{maxima}} = \frac{5V - 1,8V}{0,005A} = 640\Omega$
- Rango =  $320\Omega - 640\Omega$

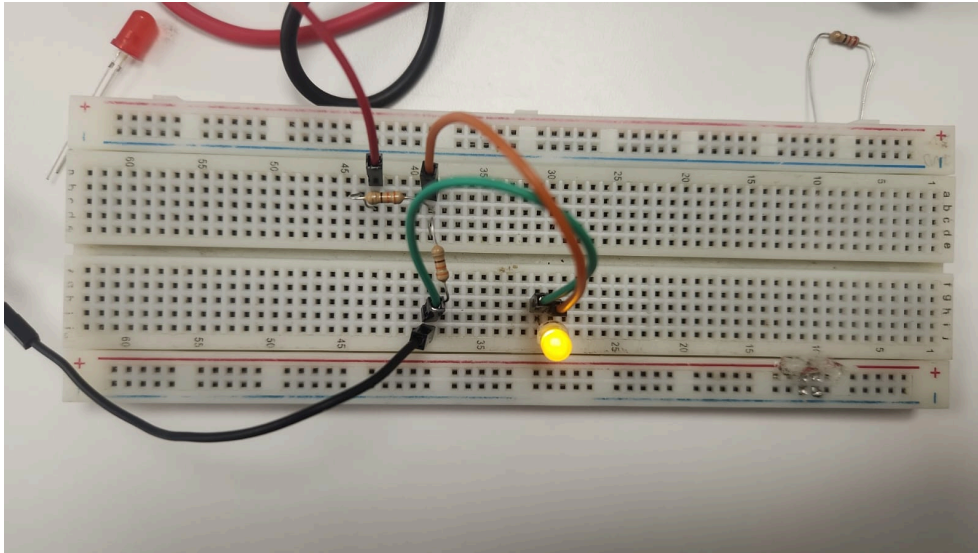
2) LED VERDE Y AMARILLO ( $V_{\text{lumbral}} = 2V$ )

- $R_{\text{minima}} = \frac{5V - 2V}{0,01A} = 300\Omega$
- $R_{\text{maxima}} = \frac{5V - 2V}{0,005A} = 600\Omega$
- Rango =  $300\Omega - 600\Omega$

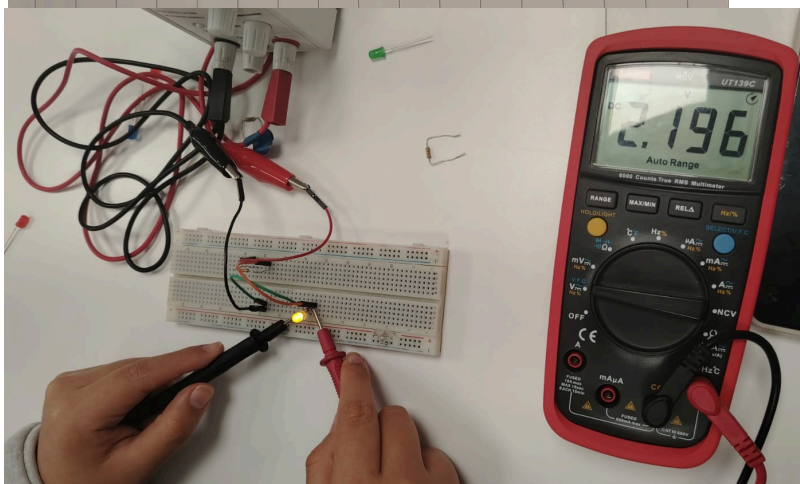
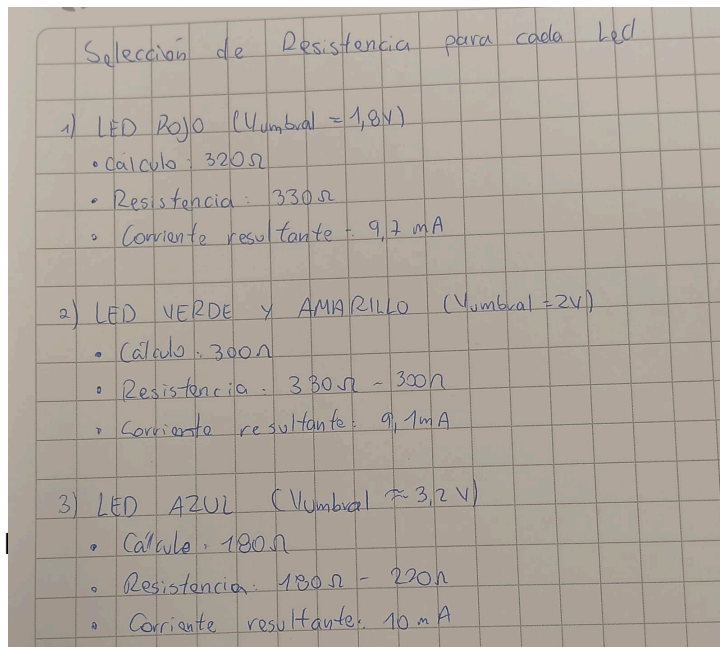
3) LED AZUL ( $V_{\text{lumbral}} = 3 \sim 3,2$ ) usamos  $3,2V$

- $R_{\text{minima}} = \frac{5V - 3,2V}{0,01A} = 180\Omega$
- $R_{\text{maxima}} = \frac{5V - 3,2V}{0,005A} = 360\Omega$
- Rango =  $180\Omega - 360\Omega$





Calculo de seleccion de resistencia para cada LED





Error absoluto ( $E_a$ )

$$\text{Error abs} = |V_{\text{medida}} - V_{\text{teórico}}|$$

$$E_{\text{abs}} = 2,196\text{V} - 2,0\text{V} = \boxed{0,196\text{V}} \leftarrow E_{\text{abs}}$$

$$\text{Error Relativo} = \frac{E_a}{V_{\text{teórico}}} = \frac{0,196\text{V}}{2,0\text{V}} = \boxed{0,098} \leftarrow E_r$$

Percentage error ( $\%E$ )

$$\%E = E_r \times 100\%$$

$$\%E = 0,098 \times 100\% = \boxed{9,8\%} \leftarrow \%E$$



### EJERCICIO 3:

## Manejo de sensor LM35

